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2027 CYCLE

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CUET UG

Mathematics

Probability

Conditional Probability, Independence and Bayes' Theorem · Section A 5 + B1 VI · CUET UG 2027 Cycle Notes

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Aligned to the latest official syllabus · Concept-first explanations

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Syllabus Alignment

Sub-topic	Official syllabus pointer (NTA CUET(UG) 2026 syllabus — Mathematics/Applied Mathematics (319), the current 2027 baseline)	Skills the questions test
Simple probability foundations	"Simple probability." (Section A Unit 5, compulsory)	Counting sample spaces fast; complements and boundary values without slips.
Conditional probability and multiplication	"Conditional probability, Multiplications theorem on probability, independent events" (Section B1 Mathematics Unit VI)	Shrinking the universe on demand; chaining probabilities along a tree.
Total probability and Bayes' theorem	"total probability, Baye's theorem" (Section B1 Mathematics Unit VI)	Weighing doors, adding the tree, then flipping the arrow — effect to cause.
Random variables and distributions (bridge)	"Random variables and its probability distribution, mean and variance of random variables" (Section B1 Mathematics Unit VI, closing part — full treatment continues in the Probability Distributions notes)	Building a distribution table from a story; reading its mean and variance.

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EXAM PATTERN

CUET UG pattern (NTA official information bulletin, 2026 cycle — the current 2027 baseline): each subject paper carries **50 compulsory questions** in **60 minutes**, with +5 for every correct answer and −1 for every incorrect answer. The 2026 syllabus lists simple probability in **Section A Unit 5** and this chapter's full toolkit — conditioning, independence, total probability, Bayes — in **Section B1 Unit VI**, so the material serves every candidate who sits the Mathematics paper.

Foundations — the Counting That Feeds Everything

Probability starts as careful counting: $P(E) = \frac{\text{favourable outcomes}}{\text{total outcomes}}$, always between 0 and 1, with the certain event at 1 and the impossible event at 0. The three standard sample spaces are worth knowing at reflex speed: one die gives 6 outcomes, two dice give $6 \times 6 = 36$ **ordered** pairs, a deck gives 52 cards in 4 suits of 13. The complement rule $P(\text{not } E) = 1 - P(E)$ is the single most useful line in the chapter — "at least one" questions fall to it instantly, and it converts many ugly direct counts into one subtraction.

- **Mutually exclusive** events cannot happen together: $P(A \cap B) = 0$, so $P(A \cup B) = P(A) + P(B)$.

- **Exhaustive** events together cover the whole sample space: their probabilities sum to 1 — the fact that powers Bayes' theorem later.
- **Addition in general** needs the overlap subtracted: $P(A \cup B) = P(A) + P(B) - P(A \cap B)$. Forgetting the overlap is the classic slip; a "probability greater than 1" answer is the alarm bell that catches it.

WORKED EXAMPLE — TWO DICE, ONE CLEAR HEAD

Two fair dice are rolled. Find the probability that the sum is 8.

Solution. Sample space: 36 ordered pairs. Favourable: $(2,6), (3,5), (4,4), (5,3), (6,2)$ — five pairs, so $P = \frac{5}{36}$. Two classic slips to dodge: counting $(4,4)$ only once is **correct** (it is one outcome), and forgetting $(5,3)$ vs $(3,5)$ are **different** pairs is not. Ordered-pair counting settles both. Complement flavour: the probability the sum is **not** 8 is $1 - \frac{5}{36} = \frac{31}{36}$ — one line, no recount.

Conditional Probability — Shrinking the Universe

$P(A \mid B)$ reads "the probability of A **given that** B happened", and the phrase "given that" physically shrinks your universe: everything outside B is deleted, and B becomes the new denominator.

THE CONDITIONING IDENTITY

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)} \quad \text{with } P(B) > 0$$

$$\text{Card reflex:} \quad P(\text{king} \mid \text{red}) = \frac{2}{26} = \frac{1}{13} \quad (26 \text{ red cards, } 2 \text{ red kings})$$

- **Read the words before the numbers.** "A card is red; find the probability it is a king" and "a card is a king; find the probability it is red" have **different** universes (26 vs 4) — and different answers ($\frac{1}{13}$ vs $\frac{2}{4} = \frac{1}{2}$).
- **Conditioning is not symmetry.** $P(A \mid B) \neq P(B \mid A)$ in general; confusing the two arrow directions is the single most common error in this chapter, and option lists are built around it.

The Multiplication Theorem and Independence

Rearranging the conditioning identity gives the **multiplication theorem**: $P(A \cap B) = P(B) \cdot P(A \mid B) = P(A) \cdot P(B \mid A)$. For chains of events it extends along a tree: multiply the branch probabilities along the path. **Independent** events are the special case where conditioning does nothing — knowing B occurred tells you nothing about A: $P(A \mid B) = P(A)$, equivalently $P(A \cap B) = P(A) \cdot P(B)$.

- **Independence and mutual exclusivity are different beasts.** Disjoint events with positive probability are **deeply dependent** — one happening kills the other. Independent events happily overlap. Any option pairing the two ideas is bait.

- **With replacement vs without.** Drawing two cards: with replacement, the second draw still sees 52 cards and probabilities multiply cleanly; without replacement, the deck has 51 cards and conditioning shifts every number. Read those two words in the question stem every time.
- **"At least one" = 1 − none.** For n independent tries each succeeding with probability p: $P(\text{at least one success}) = 1 - (1-p)^n$. One subtraction replaces a page of cases.

WORKED EXAMPLE — A THREE-BRANCH CHAIN

A bag holds 5 red and 3 black balls. Two balls are drawn one after the other **without replacement**. Find the probability that both are red.

Solution. First draw: $P(R_1) = \frac{5}{8}$. Given R_1, the bag now holds 4 red and 3 black: $P(R_2 \mid R_1) = \frac{4}{7}$. Multiply along the branch:

$$P(R_1 \cap R_2) = \frac{5}{8} \cdot \frac{4}{7} = \frac{5}{14}.$$

The conditioning did the heavy lifting — the second denominator is 7, not 8, because the universe shrank. "With replacement" would have given $\frac{5}{8} \cdot \frac{5}{8} = \frac{25}{64}$; the two-word difference in the stem changes the answer.

Total Probability and Bayes' Theorem

Bayes questions all share one story shape: several **sources** (machines, boxes, classes, buses) each produce outcomes with their own rates, you observe an **effect** (a defective bulb, a red ball, a lost article), and you must infer which source it came from. Two formulas cooperate. **Total probability** adds up the doors: if $E_1, \ldots, E_n$ partition the possibilities, $P(A) = \sum_i P(E_i) \cdot P(A \mid E_i)$. **Bayes' theorem** then flips the arrow — you saw the effect, you want the cause:

THE TWO-LINE BAYES RITUAL

$$P(A) = \sum_i P(E_i) \cdot P(A \mid E_i) \quad \text{(the whole tree, added)}$$

$$P(E_i \mid A) = \frac{P(E_i) \cdot P(A \mid E_i)}{\sum_j P(E_j) \cdot P(A \mid E_j)} \quad \text{(one door on top, all doors below)}$$

WORKED EXAMPLE — THE DEFECTIVE BULB CLASSIC

Machines A, B and C produce 25%, 35% and 40% of a factory's bulbs. Defect rates: 5%, 4% and 2% respectively. A bulb is drawn at random and found defective. What is the probability it came from machine A?

Solution. Doors: $P(A) = 0.25$, $P(B) = 0.35$, $P(C) = 0.40$ (they sum to 1 — always check). Effect D = defective:

$$P(D) = 0.25 \cdot 0.05 + 0.35 \cdot 0.04 + 0.40 \cdot 0.02 = 0.0125 + 0.0140 + 0.0080 = 0.0345.$$

Bayes flips the arrow:

$$P(A \mid D) = \frac{0.0125}{0.0345} = \frac{25}{69} \approx 0.362.$$

Machine A produces a quarter of the bulbs but over a third of the defects — the high defect rate lifted its share. The fraction $\frac{25}{69}$ (multiply the three door products by 1000: $\frac{125}{345}$) is the clean exact form options prefer.

EVERYDAY EXAMPLE — THE HOSPITAL TEST

A rare-disease test is 99% accurate, and the disease affects 1 in 1000 people. You test positive — are you probably sick? Intuition screams yes; Bayes says compute the doors: true positives $0.001 \cdot 0.99 \approx 0.001$, false positives $0.999 \cdot 0.01 \approx 0.01$. The false-positive door is **ten times bigger**, so most positive results are false alarms. That reversal of intuition — the effect felt certain, the cause turned out unlikely — is exactly what Bayes' theorem measures, and why every syllabus insists on it: it is arithmetic that corrects your gut.

Random Variables — the Bridge to Distributions

A **random variable** X assigns a number to each outcome of an experiment — number of heads in two tosses, number of defective bulbs in a sample. Its **probability distribution** is a table listing every value of X with its probability, and two facts police every table: each probability sits in $[0, 1]$, and the whole column sums to exactly 1. From the table, the **mean** is $\Sigma x_i p_i$ (the long-run average) and the **variance** is $\Sigma x_i^2 p_i - \mu^2$ where μ is the mean. Work a small one now — it is the doorway to the next chapter's binomial, Poisson and normal distributions.

WORKED EXAMPLE — TWO COINS, ONE TABLE

Two fair coins are tossed; X = number of heads. Build the distribution and find the mean and variance.

Solution. Outcomes: HH, HT, TH, TT. Values: $X = 0, 1, 2$ with probabilities $\frac{1}{4}, \frac{2}{4}, \frac{1}{4}$. Check: the column sums to 1 ✓. Mean:

$$\mu = 0 \cdot \frac{1}{4} + 1 \cdot \frac{2}{4} + 2 \cdot \frac{1}{4} = 1.$$

Variance: $\Sigma x_i^2 p_i = 0 + 1 \cdot \frac{1}{2} + 4 \cdot \frac{1}{4} = \frac{3}{2}$, so

$$\sigma^2 = \frac{3}{2} - 1^2 = \frac{1}{2}.$$

Mean 1 and variance $\frac{1}{2}$ — the table carries everything; no formula is being taken on faith.



Figure 1: The conditioning toolkit: four laws, the full Bayes tree with the defective-bulb numbers, and the traps each law hides.

Formula Cheatsheet — the Whole Chapter on One Page

FORMULA CHEATSHEET — PROBABILITY AND BAYES

Formula / rule	Letters mean	Fire it when
$P(E) = \frac{\text{favourable}}{\text{total}}$	always in $[0, 1]$	direct counting
$P(\text{not } E) = 1 - P(E)$	the rescue move	"at least one", "not"
$P(A \cup B) = P(A) + P(B) - P(A \cap B)$	overlap subtracted	two-event unions
Exclusive: overlap 0, plain add	no subtraction	disjoint events
$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$	universe shrinks to B	"given that" stems
Chain: $P(A \cap B) = P(B) \cdot P(A \mid B)$	multiply along the tree	sequential draws
Independent: $P(A \cap B) = P(A) \cdot P(B)$	no information flows	independence checks
"At least one" $= 1 - P(\text{none}) = 1 - (1 - p)^n$	p per try, n tries	repeated independent tries

Formula / rule	Letters mean	Fire it when
Total: $P(A) = \Sigma P(E_i) \cdot P(A \mid E_i)$	doors partition the space	effect through many doors
Bayes: $P(E_i \mid A) = \frac{P(E_i) \cdot P(A \mid E_i)}{\Sigma_j P(E_j) \cdot P(A \mid E_j)}$	one door on top, all below	effect -> cause
Distribution: values with probabilities, column sums to 1	policing check	table questions
Mean $\mu = \Sigma x_i p_i$; variance $\Sigma x_i^2 p_i - \mu^2$	long-run average and spread	distribution tables

Common Errors and Special Cases

COMMON ERRORS AND SPECIAL CASES — PROBABILITY

The mistake	What is right	The trap it sets
Answering $P(B \mid A)$ for $P(A \mid B)$	The word "given" names the universe	Mirror-image option pairs
Treating exclusive as independent	Disjoint events with $P > 0$ are dependent	"independent" attached to exclusive sets
Forgetting the overlap in unions	Always subtract $P(A \cap B)$	Answer above 1 — the built-in alarm
Second draw unchanged without replacement	Universe shrank: 51 cards, 4/7 bags	Recycled first-draw denominators
Bayes denominator = one door	The sum of ALL doors	Options = one-door fraction, plausible but wrong
Chaining "at least one" by cases	Complement: $1 - (1 - p)^n$	Case-counting that leaks a branch
Distribution column not summing to 1	Policing check before computing	Tables with a sneaky missing outcome
Variance computed as $\Sigma x_i^2 p_i$ alone	Subtract μ^2	Inflated-variance numeric options

Sixty-Second Self-Test

Statement	Verdict
If events are mutually exclusive, they are independent.	False — one kills the other: maximal dependence
$P(A \mid B) = P(B \mid A)$ in general.	False — the arrow has a direction
With replacement, draw-two probabilities stay equal.	True — the deck resets
Bayes needs the door weights to sum to 1.	True — the partition check comes first
"At least one six in 2 dice" $= 1 - \left(\frac{5}{6}\right)^2 = \frac{11}{36}$.	True — complement route
A distribution table may sum to 0.9.	False — exactly 1, always

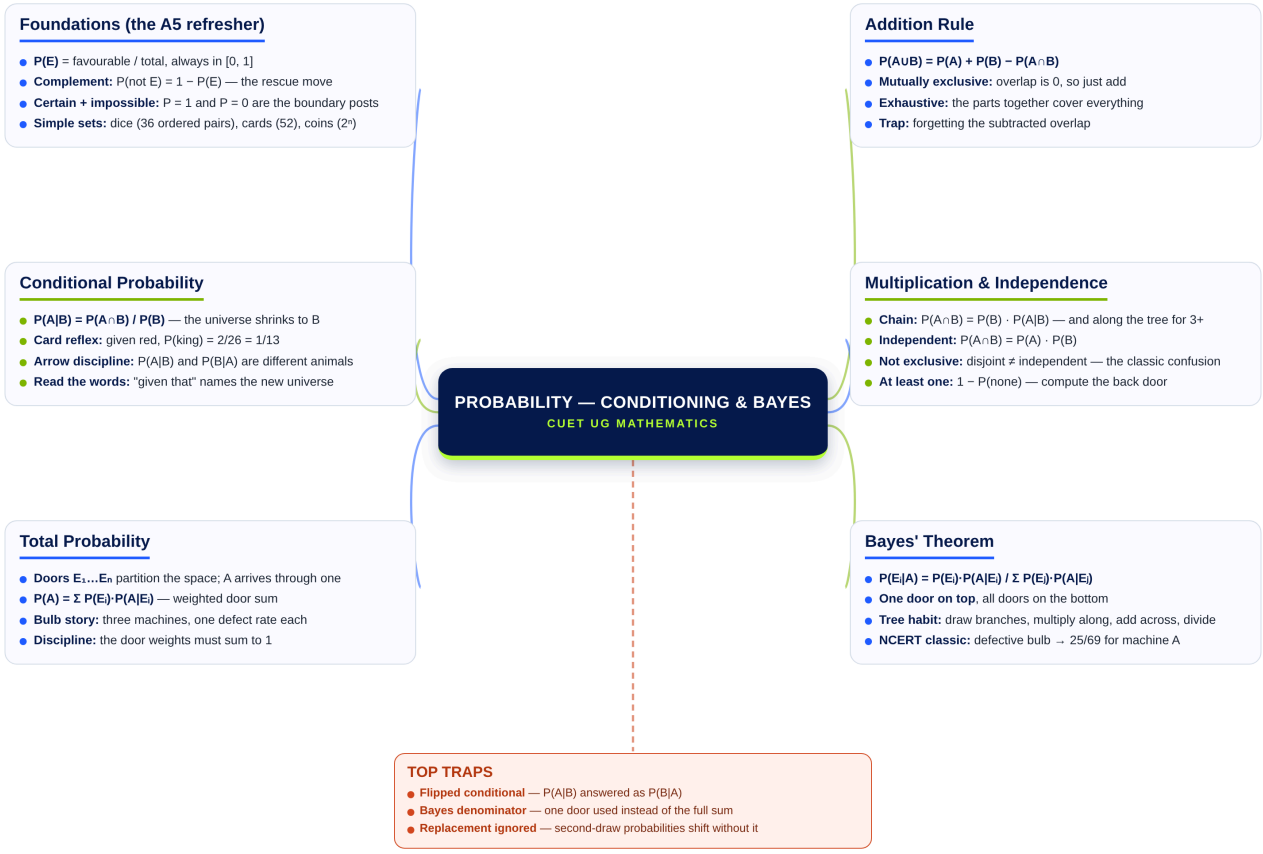


Figure 2: The chapter in one look: foundations, conditioning, independence, total probability, Bayes — and the traps between them.

Practice Questions — CUET UG Pattern (10 MCQs)

Ten questions in the CUET UG style. Each correct answer earns +5; each incorrect one costs −1. Read "given that" and "without replacement" first — those two phrases decide most of this set.

Q1. Two fair dice are rolled. The probability that the sum is 8 is:

(A) $\frac{1}{6}$ **(B)** $\frac{5}{36}$
(C) $\frac{1}{9}$ **(D)** $\frac{7}{36}$

Q2. A card is drawn from a well-shuffled deck. Given that it is red, the probability that it is a king is:

(A) $\frac{1}{26}$ **(B)** $\frac{1}{2}$
(C) $\frac{2}{13}$ **(D)** $\frac{1}{13}$

Q3. If $P(A) = 0.5$, $P(B) = 0.4$ and $P(A \cap B) = 0.2$, then $P(A \cup B)$ equals:

(A) 0.9 **(B)** 0.3
(C) 0.7 **(D)** 1.1

Q4. If A and B are independent with $P(A) = 0.6$ and $P(B) = 0.5$, then $P(A \cap B)$ equals:

(A) 0.3 **(B)** 0.1
(C) 0.55 **(D)** 0.8

Q5. A bag has 5 red and 3 black balls. Two are drawn without replacement. The probability both are red is:

(A) $\frac{25}{64}$ **(B)** $\frac{5}{8}$
(C) $\frac{5}{14}$ **(D)** $\frac{3}{14}$

Q6. A fair coin is tossed 3 times. The probability of at least one head is:

(A) $\frac{1}{2}$ **(B)** $\frac{7}{8}$
(C) $\frac{3}{8}$ **(D)** $\frac{5}{8}$

Q7. Machines A, B, C make 25%, 35%, 40% of bulbs with defect rates 5%, 4%, 2%. A bulb is defective. The probability it came from A is:

(A) $\frac{25}{69}$ **(B)** $\frac{1}{4}$
(C) $\frac{5}{23}$ **(D)** $\frac{69}{100}$

Q8. Two fair coins are tossed and X = number of heads. The mean of X is:

(A) 1/2 **(B)** 3/4
(C) 1 **(D)** 3/2

Q9. Two dice are rolled. Given that the sum is 8, the probability that the first die shows 4 is:

(A) $\frac{1}{6}$ **(B)** $\frac{1}{5}$
(C) $\frac{1}{36}$ **(D)** $\frac{5}{36}$

Q10. For the same X (number of heads in two tosses), the variance of X is:

(A) 1 **(B)** 3/4
(C) 1/4 **(D)** 1/2

Detailed Solutions

Sol. Q1. Favourable ordered pairs: $(2,6), (3,5), (4,4), (5,3), (6,2)$ — five of 36. $P = \frac{5}{36}$. Count $(4,4)$ once, count $(3,5)$ and $(5,3)$ separately.

Answer: (B)

Sol. Q2. "Given red" shrinks the deck to 26 cards containing exactly 2 red kings: $\frac{2}{26} = \frac{1}{13}$. Option (B) answers the flipped question $P(\text{red} \mid \text{king}) = \frac{2}{4}$ — the arrow trap in its natural habitat.

Answer: (D)

Sol. Q3. Addition rule: $0.5 + 0.4 - 0.2 = 0.7$. Option (A) forgot the overlap and exceeded sanity — the alarm the rule exists to ring.

Answer: (C)

Sol. Q4. Independence means multiply: $0.6 \cdot 0.5 = 0.3$. Option (B) subtracted — the exclusive-event reflex applied where it does not belong.

Answer: (A)

Sol. Q5. Chain with shrunken universe: $\frac{5}{8} \cdot \frac{4}{7} = \frac{20}{56} = \frac{5}{14}$. Option (A) is the with-replacement answer $\frac{25}{64}$ — the two-word stem difference, priced at one full question.

Answer: (C)

Sol. Q6. Complement route: $P(\text{no heads}) = \left(\frac{1}{2}\right)^3 = \frac{1}{8}$, so $P(\text{at least one}) = 1 - \frac{1}{8} = \frac{7}{8}$. Option (C) is "exactly one head" $\left(\frac{3}{8}\right)$ — a different event wearing similar words.

Answer: (B)

Sol. Q7. Doors: $0.25 \cdot 0.05 = 0.0125$, $0.35 \cdot 0.04 = 0.0140$, $0.40 \cdot 0.02 = 0.0080$; total $P(D) = 0.0345$. Bayes: $\frac{0.0125}{0.0345} = \frac{125}{345} = \frac{25}{69}$. Option (C) used only machine A's door as the denominator.

Answer: (A)

Sol. Q8. Distribution: $0, 1, 2$ with $\frac{1}{4}, \frac{1}{2}, \frac{1}{4}$. Mean $= 0 + \frac{1}{2} + \frac{1}{2} = 1$. The symmetric table makes the mean obvious — half of two tosses.

Answer: (C)

Sol. Q9. Condition on sum 8: the universe is the 5 pairs $(2,6), (3,5), (4,4), (5,3), (6,2)$. Exactly one has first die 4: $\frac{1}{5}$. The unconditioned $\frac{1}{6}$ and $\frac{5}{36}$ are the pre-conditioning lures.

Answer: (B)

Sol. Q10. Variance: $\Sigma x_i^2 p_i = 0 + 1 \cdot \frac{1}{2} + 4 \cdot \frac{1}{4} = \frac{3}{2}$; minus $\mu^2 = 1$: $\sigma^2 = \frac{1}{2}$. Option (A) skipped the μ^2 subtraction — the exact error the errors-box warns about.

Answer: (D)

Wrong-Answer Clinic

If you picked this	The thinking slip	The lasting fix
(A) in Q2	Answered $P(\text{red} \mid \text{king})$ instead	Underline the "given" clause; that set is the new denominator
(A) in Q3	Added without subtracting the overlap	Union above 1 or overlap ignored — run the subtraction reflex
(A) in Q5	Reused the 8-ball bag for the second draw	"Without replacement" = shrink the bag, then condition
(C) in Q6	Counted "exactly one"	"At least one" = complement of "none" — always
(C) in Q7	One door in the Bayes denominator	The bottom of Bayes is the WHOLE tree — sum all doors
(A) in Q10	Variance without the μ^2 correction	$\Sigma x_i^2 p_i - \mu^2$: two-term recipe, both terms

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